Python Notes for Java Learners

🌟 What is Python?

Python is a **high-level, easy-to-read programming language** created by **Guido van Rossum**.

It is famous for:

- ✅ Simple syntax
- ✅ Easy to learn
- ✅ Less code compared to Java
- ✅ Powerful libraries
- ✅ Used in AI, Web Development, Automation, Data Science, Cyber Security, etc.

🔥 Simple Python Program

```
print("Hello World")
```

Output

```
Hello World
```

☕ Java vs 🐍 Python

Feature	Java	Python
Typing	Static Typing	Dynamic Typing
Variable Declaration	Datatype required	No datatype needed
Execution Type	Compiled + Interpreted	Interpreted
Platform Independent	Yes (JVM)	Yes
Syntax	More Complex	Simple & Clean
Code Length	More	Less

Feature	Java	Python
Speed	Faster	Slightly Slower
Memory Usage	More	Less
Beginner Friendly	Medium	Very Easy
Security	More Secure	Less Secure compared to Java
Multithreading	Strong Support	Limited compared to Java
Development Speed	Slower	Faster
Popular Use Cases	Enterprise Apps, Android Apps, Banking Systems	AI, Data Science, Automation, Web Development
File Extension	`.java`	`.py`

⚙️ How Java Works

1️⃣ Java code is first compiled into **Bytecode**.

2️⃣ JVM (Java Virtual Machine) interprets the bytecode.

👉 That is why Java is called:

- **Compiled + Interpreted Language**
- **Platform Independent Language**

Java Flow

```
Java Code → Compiler → Bytecode → JVM → Output
```

⚙️ How Python Works

Python code is directly executed by the Python Interpreter.

Python Flow

```
Python Code → Interpreter → Output
```

👉 Python is mainly called an:

- **Interpreted Language**
- **High-Level Language**

🌍 Platform Independence

☕ Java

Java uses JVM.

👉 "Write Once, Run Anywhere"

A Java program can run on:

- Windows
- Linux
- Mac

if JVM is installed.

🐍 Python

Python is also platform independent.

If Python Interpreter is installed, the same program can run on different operating systems.

🔐 Security Comparison

☕ Java

- More secure
- Strong memory management
- Better for enterprise and banking systems

🐍 Python

- Easier and flexible
- Slightly less secure compared to Java

- Mostly preferred for rapid development

🚀 Where Java is Used

✅ Android Applications

✅ Banking Systems

✅ Enterprise Applications

✅ Desktop Applications

✅ Backend Systems

✅ Cloud Applications

🤖 Where Python is Used

✅ Artificial Intelligence (AI)

✅ Machine Learning

✅ Data Science

✅ Automation

✅ Web Development

✅ Cyber Security

✅ Scripting

✅ Game Development

🔷 Variable Declaration

☕ Java

```
int age = 21;
String name = "Rahul";
```

🐍 Python

```
age = 21
name = "Rahul"
```

👉 Python automatically understands the datatype.

🧠 Dynamic Typing in Python

```
x = 10
print(x)

x = "Python"
print(x)
```

Output

```
10
Python
```

👉 Same variable can store different datatype values.

📘 What is a String?

A **String** is a sequence of characters.

🔤 Creating Strings

☕ Java

```
String name = "Python";
```

🐍 Python

```
name = "Python"
```

🎨 Types of Quotes in Python

```
single = 'Hello'
double = "Hello"
triple = '''Hello'''
```

🟡 Multiline Text

☕ Java

```
String text = "Hello\nWorld";
```

🐍 Python

```
text = """
Hello
World
"""
```

👉 Python multiline strings are much easier.

🔍 Accessing Characters

☕ Java → charAt()

```
String name = "Python";
System.out.println(name.charAt(0));
```

Output

```
P
```

🐍 Python → Indexing

```
name = "Python"
print(name[0])
```

Output

```
P
```

🔁 Negative Indexing in Python

```
name = "Python"

print(name[-1])
```

Output

```
n
```

👉 Java does not support negative indexing directly.

✂️ Substring / Slicing

☕ Java → substring()

```
String text = "Programming";

System.out.println(text.substring(0, 6));
```

Output

```
Progra
```

🐍 Python → Slicing

```
text = "Programming"

print(text[0:6])
```

Output

```
Progra
```

🎯 More Slicing Examples in Python

```
text = "Programming"

print(text[:5])
print(text[3:])
print(text[::2])
```

Output

```
Progr
gramming
Pormig
```

⚡ String Operations

1️⃣ Concatenation

☕ Java

```
String first = "Hello";
String second = "World";

System.out.println(first + " " + second);
```

🐍 Python

```
first = "Hello"
second = "World"

print(first + " " + second)
```

Output

```
Hello World
```

2️⃣ Repetition

☕ Java

❌ No direct operator.

Need loops.

🐍 Python

```
print("Hi " * 3)
```

Output

```
Hi Hi Hi
```

3️⃣ Membership Checking

☕ Java

```
String text = "Python Programming";

System.out.println(text.contains("Python"));
```

🐍 Python

```
text = "Python Programming"

print("Python" in text)
```

Output

```
True
```

📚 What are Methods?

Methods are functions attached to an object.

Example:

```
text.upper()
```

Here:

- `upper()` is a method
- It belongs to the string object

📚 What are Functions?

Functions are reusable blocks of code.

Example:

```
len(text)
```

👉 `len()` is a function.

🔥 Difference Between Methods and Functions

Functions	Methods
Independent	Belong to object
Example: `len(text)`	Example: `text.upper()`
Called directly	Called using dot operator

🧪 Important String Methods in Python

1️⃣ upper()

☕ Java

```
String text = "python";

System.out.println(text.toUpperCase());
```

🐍 Python

```
text = "python"

print(text.upper())
```

Output

```
PYTHON
```

2️⃣ lower()

☕ Java

```
System.out.println(text.toLowerCase());
```

🐍 Python

```
print(text.lower())
```

3️⃣ length vs len()

☕ Java

```
String name = "Python";

System.out.println(name.length());
```

🐍 Python

```
name = "Python"

print(len(name))
```

Output

```
6
```

4️⃣ replace()

☕ Java

```
String text = "I like Java";

System.out.println(text.replace("Java", "Python"));
```

🐍 Python

```
text = "I like Java"

print(text.replace("Java", "Python"))
```

Output

```
I like Python
```

5️⃣ split()

☕ Java

```
String fruits = "apple,banana,mango";

System.out.println(Arrays.toString(fruits.split(",")));
```

🐍 Python

```
fruits = "apple,banana,mango"

print(fruits.split(","))
```

Output

```
['apple', 'banana', 'mango']
```

6️⃣ join()

☕ Java

```
String.join(" - ", arr)
```

🐍 Python

```
items = ['Python', 'Java', 'C++']

print(" - ".join(items))
```

Output

```
Python - Java - C++
```

7️⃣ find()

☕ Java

```
String text = "Python";

System.out.println(text.indexOf("th"));
```

🐍 Python

```
text = "Python"

print(text.find("th"))
```

Output

```
2
```

8️⃣ count()

☕ Java

❌ No direct method.

Need loops.

🐍 Python

```
text = "apple apple mango"

print(text.count("apple"))
```

Output

```
2
```

🧵 Escape Characters

Escape Character	Meaning
`\n`	New Line
`\t`	Tab Space
`\\`	Backslash
`\"`	Double Quote

Example

```
print("Hello\nWorld")
```

Output

```
Hello
World
```

🧊 Strings are Immutable

👉 Immutable means cannot be changed directly.

☕ Java

Strings are immutable.

🐍 Python

Strings are also immutable.

❌ Wrong

```
text = "Python"

# text[0] = 'J'
```

✅ Correct

```
text = "Python"

text = "J" + text[1:]

print(text)
```

Output

```
Jython
```

🎯 String Formatting

☕ Java

```
String name = "Rahul";
int age = 21;

System.out.println("My name is " + name + " and age is " + age);
```

🐍 Python → f-String

```
name = "Rahul"
age = 21

print(f"My name is {name} and age is {age}")
```

Output

```
My name is Rahul and age is 21
```

🚀 Complete Example Program

```
name = input("Enter your name: ")

print("Uppercase:", name.upper())
print("Lowercase:", name.lower())
print("Length:", len(name))
print("First Character:", name[0])
print("First 3 Letters:", name[:3])
```

🎓 Final Summary

Python Advantages Over Java for Strings

✅ Less code

✅ Easier syntax

✅ Powerful slicing

✅ Negative indexing

✅ Built-in operations

✅ Easier multiline strings

🧠 Quick Revision Table

Concept	Java	Python
Access Character	`charAt()`	`[]` indexing
Substring	`substring()`	slicing `[:]`
Length	`length()`	`len()`
Uppercase	`toUpperCase()`	`upper()`
Lowercase	`toLowerCase()`	`lower()`
Contains	`contains()`	`in`
Find Position	`indexOf()`	`find()`
Multiline String	`\n`	Triple Quotes
Repeat String	Loops	`*` operator

🌟 Conclusion

If you already know Java, learning Python becomes much easier.

Python removes a lot of extra syntax and makes programming:

- Faster
- Cleaner
- Easier to understand

That is why Python is one of the most popular programming languages today.